TOP-N RECOMMENDER SYSTEM VIA MATRIX COMPLETION

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INTRODUCTION

Recommender Systems: Information filtering systems that suggest relevant items (movies, products, music) to users based on preferences.

Recommender Systems are widely used in platforms like Netflix, Amazon, and Spotify.

Matrix Completion: A mathematical technique to fill missing entries in a user–item rating matrix.

PROBLEM STATEMENT

User–item rating matrix is sparse.

Need to predict missing entries for accurate recommendations.

Direct rank minimization is NP-hard.

Nuclear norm over-penalizes large singular values.

LogDet function is proposed as a better surrogate for rank.

OPTIMIZATION PROBLEM

- Represent ratings as a matrix $M \varepsilon R^{m*n}$.
- Observed entries defined by the index set Ω.

- Rank Minimization Problem:

$$\min_{x} \operatorname{rank}(X) \text{ s.t } P_{\Omega}(X) = P_{\Omega}(M), X \geq 0$$

M: observed user–item rating matrix

Ω: set of observed entries

- Problem is **NP-hard** → cannot be solved efficiently for large datasets.
- Motivates the use of **surrogate functions** (Nuclear Norm, LogDet).

Objective Equation :

$$\min_{X \geq 0} \log \det((X^T X)^{\frac{1}{2}} + I) \quad + \frac{\mu}{2} \parallel I_\Omega \odot (X - M) \parallel_F^2$$

- **X** → the **completed user–item rating matrix** we are trying to learn
- **M** → the **observed ratings matrix** (sparse, from dataset).
- **Ω**→ the set of indices where ratings are known (observed entries).
- I_Ω→ indicator matrix: 1 if an entry is observed, 0 if missing.
- $\odot$→ elementwise multiplication (Hadamard product).
- $\parallel \ \parallel_F$ → Frobenius norm (like Euclidean norm for matrices).
- **μ**→ penalty parameter (controls how strongly we enforce data-fitting to observed ratings).
- $\log \det((x^T x)^{1/2} + I)$→ the **LogDet surrogate**, which approximates the rank of X in a smoother, more stable way.

AUGMENTED LANGRAGEIAN

- After introducing the auxiliary variable Y (for nonnegativity constraint) and dual variable Z (for X=W), the augmented Lagrangian is:

$$\mathrm{L}_{\mu}(\mathrm{X}, Y, \mathrm{Z}) = \sum \log\big(1 + \sigma_i(x)\big) + \frac{\mu}{2}\left\| x - \mathrm{Y} + \frac{z}{\mu} \right\|_F^2 + l_{R_+}(Y)$$

$$\log\det((\mathrm{X}^{\mathrm{T}}\mathrm{X})^{1/2} + \mathrm{I}) \quad \rightarrow \quad \sum \log(1 + \sigma_{\mathrm{i}}(\mathrm{X}))$$

Update Equations

<u>X update:</u>

$$\mathrm{X}^{t+1} = \arg\min_{\mathrm{X}} \log\det((\mathrm{X}^{\mathrm{T}}\mathrm{X})^{1/2} + \mathrm{I}) + \frac{\mu_t}{2} \| \mathrm{X} - (\mathrm{Y}^t - \frac{\mathrm{Z}^t}{\mu_t}) \|_F^2$$

$$D = \left(\mathrm{Y}^t - \frac{\mathrm{Z}^t}{\mu_t}\right)$$

$$\rho = \frac{\mu_t}{2}$$

- SVD of the center

$$\mathrm{D} = \mathrm{U}\sum\mathrm{V}^{\mathrm{T}},\ \forall\sigma_D \in \mathrm{diag}(\Sigma)$$

- For each σ_D, solve the 1-D prox:

$$\mathrm{s}^* = arg\ \min_{\mathrm{s}\geq 0} log(1+s) + \rho(s-\sigma_{\mathrm{D}})^2$$

- First-Order condition

$$\frac{1}{1+s} + 2\rho(s-\sigma_D) = 0$$

$$s^2 + (1-\sigma_D)s + \left(\frac{1}{2\rho} - \sigma_D\right) = 0$$

- Choose s^*

$$Discriminant \ \triangle = (1-\sigma_D)^2 - 4(\frac{1}{2\rho} - \sigma_D)$$

$$Roots \ s_{1,2} = \frac{-(1-\sigma_D) \pm \sqrt{\triangle}}{2}$$

keep the nonnegative root if it decrease otherwise set s to 0

- Reconstruct X

$$X^{t+1} = U.diag(s^*).V^T$$

- Restore Observed ratings

$$X_\Omega^{t+1} = M_\Omega \ (i.e, set \ X^{t+1}(I_\Omega = 1) = M(I_\Omega = 1))$$

Y Update :

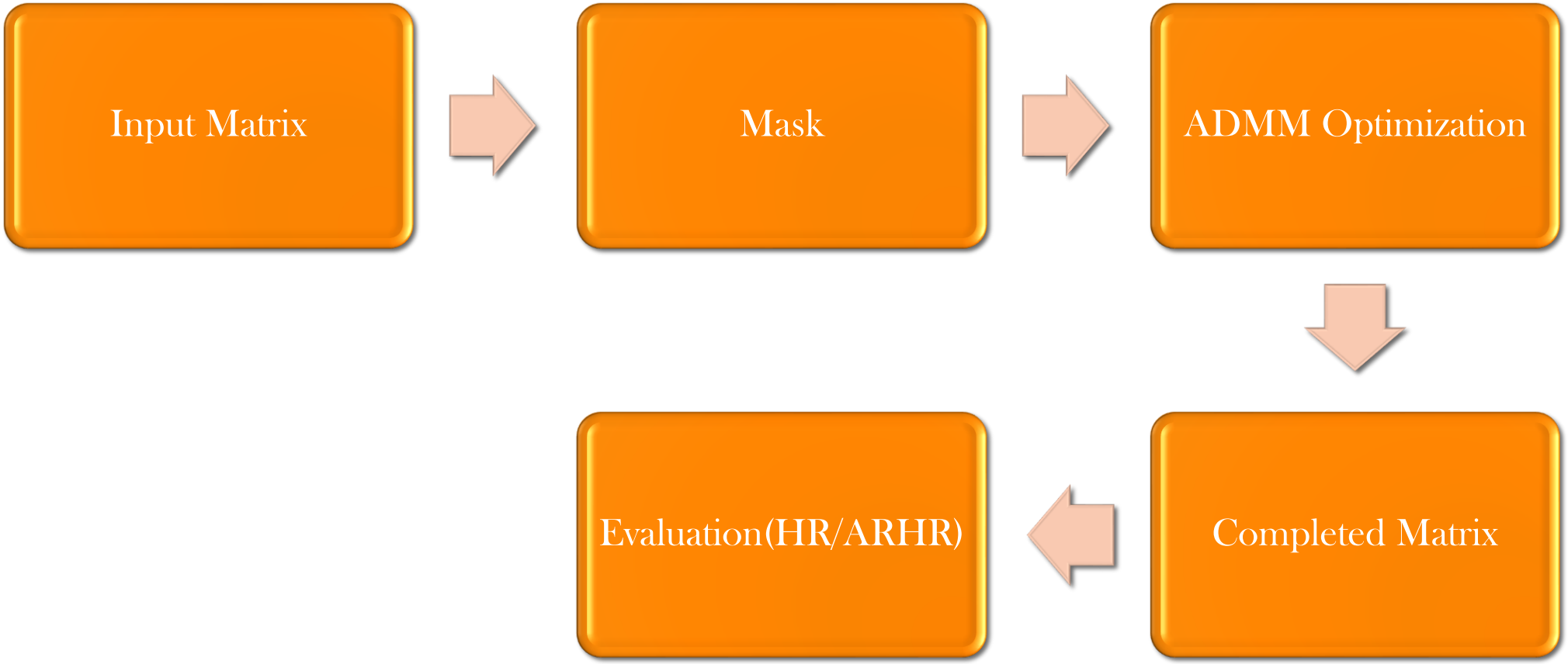
$$Y^{t+1} = \arg\min_{Y} l_{R_+}(y) + \frac{\mu_t}{2}\left\|x^{t+1} - y + \frac{z^t}{\mu_t}\right\|_F^2$$

$$Y^{t+1} = \max\left(X^{t+1} + \frac{Z^t}{\mu_t}, 0\right)$$

Z Update :

$$z^{t+1} = z^t + \mu_t(X^{t+1} - Y^{t+1})$$

METHODOLOGY



DATASETS DESCRIPTION

DATASET	DOMAIN	USERS	ITEMS	INTERACTIONS	DENSITY
MovieLens	Movies	943	1,682	100,000	≈6.3
LastFM	Music	1,892	17,632	92,834	≈0.28
Delicious	Bookmarks	2,078	12,096	437,000	≈0.20
BX	Books	2,078	9,300	96,000	≈0.50
Netflix	Movies	5,000	17,000	300,000	≈0.35

LINKS: HetRec 2011 | GroupLens, Book-Crossing Dataset, Netflix Prize data

PERFORMANCE METRICS

1. Hit Ratio (HR@N)

Definition: Fraction of users for whom the actual test item appears in the **Top-N recommended list**.

Formula:

$$HR@N = \frac{1}{\| U \|} \sum_{u \in U} 1\{p_u \in TopN(u)\}$$

Intuition: Measures *coverage* — how often we successfully recommend the correct item within the top-N.

2. Average Reciprocal Hit Rank (ARHR@N)

Definition: Rewards not only the presence of the test item in Top-N but also its **position** in the ranked list.

Formula:

$$ARHR@N = \frac{1}{|U|}\sum_{u\varepsilon U}\frac{1}{rank_u(p_u)}$$

Intuition: Measures ranking quality-higher scores mean the correct item is placed closer to the top of the list.

Results

- The LogDet-based Matrix Completion model achieved consistent HR and ARHR across multiple domains, demonstrating strong generalization on sparse datasets.

Dataset	HR@10	ARHR@10	μ	ρ	κ
MovieLens	0.9466	0.4197	0.06(initial)	0.06	2
LastFM	0.2573	0.1100	0.010	1	1
Delicious	0.2366	0.0943	0.010	1	1
BX(BookCrossing)	0.7365	0.2976	0.06	0.06	2
Netflix Subset	0.3597	0.0721	0.06	0.06	2

References

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